

MICHAEL QIN

SOFTWARE DEVELOPER

CONTACT

✉ michael.qin@uwaterloo.ca

🌐 michaelqin.net

☎ (613) 290-4310

in [michael-q](#)

🔗 [michaelqin3](#)

TECHNICAL SKILLS

C/C++, Python, HTML/CSS,
JavaScript, React, Bootstrap, Git,
Raspberry Pi

EDUCATION

University of Waterloo
Candidate for Bachelor of
Software Engineering
GPA: 85.8%

AWARDS

June 2018 Governor General's
Academic Bronze
Medal
Bell High School
Highest cumulative average
of graduating class (250
students)

Feb. 2018 Certificate of
Distinction
Canadian Computing
Competition
Top 25% in the Senior
division

Apr. 2018 Certificate of
Distinction
Euclid Math Contest
Top 25% of all contestants

Sept. 2015 Medal for Excellence
Conservatory Canada
Earned the highest mark in
Ontario for the Music
Theory 6 exam

PROJECTS

Reddit Bot

- Created a bot with **Python** and the **reddit API** which detects spelling errors
- Searches all reddit comments and posts a reply when mistakes are found
- Avoids duplicate replies by tracking comment IDs in a text file

Matrix Calculator Webapp

- Webapp calculator performs simple matrix operations using **React** and **JavaScript**
- Utilized forms to collect user input and arrays to parse data

Two-Player Tetris

- Used **Raspberry Pi** and **Python** to generate obstacles during gameplay
- Manipulated matrices to limit board size and rotation of pieces
- Interface through physical buttons connected with GPIO pins and a breadboard

Poker Hand Dealer

- Created a **C** program that shuffles, deals, and sorts poker hands by hand strength
- Implemented data structures, mergesort, and enumerated types to organize cards
- Takes user input for number of hands, then outputs sorted hands in a text file

WORK EXPERIENCE

Nepean School of Music Ottawa, Ontario
Private Teacher Sept. 2017 to Aug. 2018

- Taught piano in one-on-one lessons with students aged 4-34
- Corresponded with students' parents, summarizing each lesson
- Created a plan for progression for each student based on their individual needs

Spirit of Math Ottawa, Ontario
Teacher Assistant Sept. 2016 to June 2017

- Helped teach math at a challenging reach-ahead program to Grade 4-5 students
- Communicated tactfully between students, teachers, parents, and the principal

COMMUNITY INVOLVEMENT

Hack the North Waterloo
Checked in hackers and collected data to improve workshops

DECA Bell High School
Co-President

Co-president of the business club DECA in high school, competed provincially.
Responsible for running biweekly practices and communicating between DECA and teacher supervisors.

Competitive Ultimate Frisbee

Played two seasons, finished 16th team in Canada at 2018 Nationals

University Choir University of Waterloo
Section Leader

Bass singer in the Waterloo choir and helped lead sectionals